

MDM Assignment 12

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MDM Assignment 12

This are your test results

Knowledge Questions

 Interpretability and Evaluation

Status

Seen but not answered

Your score

0 / 6

0%

Response

Which of the following statements is **Right** and which are **Wrong**?

Note :

Correct answer : +1 points

Unanswered : 0 point

Incorrect answer : -0.5 point

Unanswered	Right	Wrong	
☐	☐	☐	F-Score is the harmonic mean of precision and recall
☐	☐	☐	Precision is a better evaluation metric than recall.
☐	☐	☐	The interpretability is important when the data is large.
☐	☐	☐	All machine learning models are interpretable.
☐	☐	☐	A machine learning model is interpretable when it is possible to understand its behaviour and outcome.
☐	☐	☐	ROC value falls between 0 and 1

 Supervised learning

Status

Not started

Your score

0 / 4

0%

Response

In supervised learning, which of the following statements is **Right** and which are **Wrong**?

Note :

Correct answer : +1 points

Unanswered : 0 point

Incorrect answer : -0.5 point

Unanswered	Right	Wrong	
☐	☐	☐	The same samples in the training set must be present in the testing set.
☐	☐	☐	The dataset must consist of samples from at least two classes.
☐	☐	☐	The data must be unlabeled.
☐	☐	☐	The feature vectors must have fixed lengths.

 MLDM Pipeline

Status Not started

Your score 0 / 7.5 0%

Response

You are asked to design a Machine Learning solution. You are provided with an x-rays dataset that consists of x-rays images acquired from a hospital's patient. The dataset contains 50000 grey-scale images where 6000 of them are labelled "broken bone" and the remaining are labelled "healthy bone". The goal is to design an MLDM model that classifies these x-rays images.

How would you design your machine learning pipeline?

Note :
Correct answer : +1.5 points
Unanswered : 0 point
Incorrect answer : -0.5 point

Unanswered	Right	Wrong	
☐	☐	☐	A neural network seems a good algorithm to tackle this problem.
☐	☐	☐	After training the model, we need to evaluate its performance on the test set using the Silhouette coefficient
☐	☐	☐	For a fair evaluation, a weighted macro-averaging score is more suitable.
☐	☐	☐	It is not necessary but we can use k-means to clean the data.
☐	☐	☐	If we have more data, we may use linear regression for a better result.

 Essay

Status Not started

Your score 0 / 11.5 0%

Response

In your own words, explain the three main evaluation levels of interpretability.

0 word



 Evaluation

 Evaluation

StatusNot started

Your score0 / 190%

Response

Given a dataset consisting of 610 images representing three classes, namely; **Beach**, **Desert** and **Forest**. 450 of them are used for training a neural network model. After applying this model on the testing set, it returned the following:

- From 60 images belonging to **Beach**, 12 of them were classified as **Desert** and 2 of them as **Forest**.
- From 70 images belonging to **Desert**, 9 of them were classified as **Beach** and the rest as **Desert**.
- From 30 images belonging to **Forest**, two of them were classified as **Beach** and 25 as **Forest**.

Fill out the following table (with two decimal digits after the point)

	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Desert				
Beach				
Forest				
Micro Average				
Macro Average				

 Decision Trees

Given the following dataset about customers of an insurance company. There are three attributes: income (High/Low), Married (Yes/No) and Gender (Male/Female). Each sample is labelled with whether the customer is creditworthy (Yes/No).

No.	Income	Married	Gender	Creditworthy
1	H	Y	M	N
2	H	N	F	Y
3	L	Y	M	Y
4	H	Y	M	N
5	L	N	F	N
6	H	Y	F	N
7	L	Y	M	Y

Following questions are about constructing a decision tree classifier for this dataset, which classifies new feature vectors consisting of the attributes Income, Married and Gender to a label Yes or No.

 Overall entropy

StatusNot started

Your score0 / 40%

Response

What is the overall entropy of the dataset?

☒ Best node split

Status

Not started

Your score

0 / 4

0%

Response

Which of the three possible node splits lead to the single-split decision tree? If more than one possibility leads to an optimal decision tree then check all of them.

☐ Married

☐ Income

☐ Gender

Perfect decision tree

Status

Not started

Your score

0 / 8

0%

Response

It is possible to construct a decision tree that fits the training data perfectly. Describe this tree by listing:

- all its nodes
- the attribute of each node
- the child nodes of each node
- the samples of each leaf node

0 word

Programming

☒ Feature selection

Status

Not started

Your score

0 / 2

0%

Response

Which of the following is to correct to select the feature vectors ["height" : index 2, "weight" : index 3, "age" : index 4, "city" : index 5]?

☐ dataframe.loc[:, "height", "weight", "age", "city"]

☐ dataframe.loc["height", "weight", "age", "city", :]

☐ dataframe.iloc["height", "weight", "age", "city", :]

☐ dataframe.iloc[2:5, :]

☐ dataframe.iloc["height", "weight", "age", "city"][:]

☐ dataframe.loc[2:5, :]

F1 score

Status

Not started

Your score

0 / 5

0%

Response

Create a function that calculates the F1-Score, where the input is a matrix that consists of two columns (the predicted class, the ground truth).

0 word

Macro-average precision

Status

Not started

Your score

0 / 7

0%

Response

Create a function that calculates the macro-average precision, where the input is a matrix that consists of two columns (the predicted class, the ground truth).

0 word

Micro-average precision

Status

Not started

Your score

0 / 7

0%

Response

Create a function that calculates the micro-average precision, where the input is a matrix that consists of two columns (the predicted class, the ground truth).

0 word

Upload file

Status

Not started

Your score

0 / 15

0%

Response

Implement a class that contains the following methods:

- A constructor that takes two parameters: a list of the predicted classes, and a list of the ground truth.
- "getPrecision": a function that returns the precision of the given predicted and ground truth lists.
- "getRecall": a function that returns the recall of the given predicted and ground truth lists.
- "getPrecision": a function that returns the precision of the given predicted and ground truth lists.
- "getAccuracy": a function that returns the accuracy of the given predicted and ground truth lists.
- "getF1Score": a function that returns the F1-score of the given predicted and ground truth lists.
- "printEvaluation": a function that visualizes all the evaluation metrics calculated in a form of a table.

File

No file uploaded